

2017

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# XCEPTION:

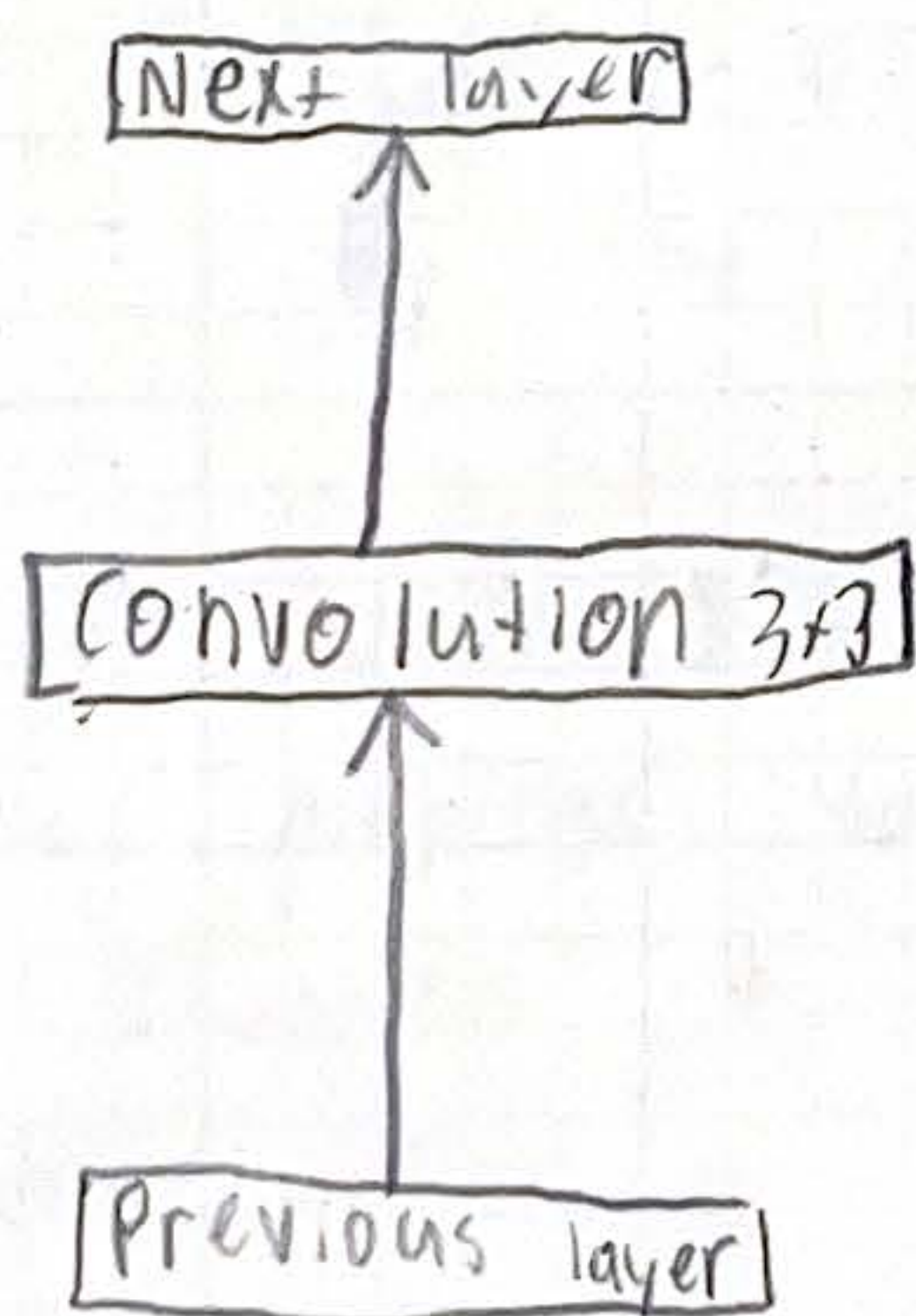
## Deep learning with Depthwise Separable Convolutions

This from my understanding is a bit more of a theoretical paper where we ask what does convolution do How can we improve it?

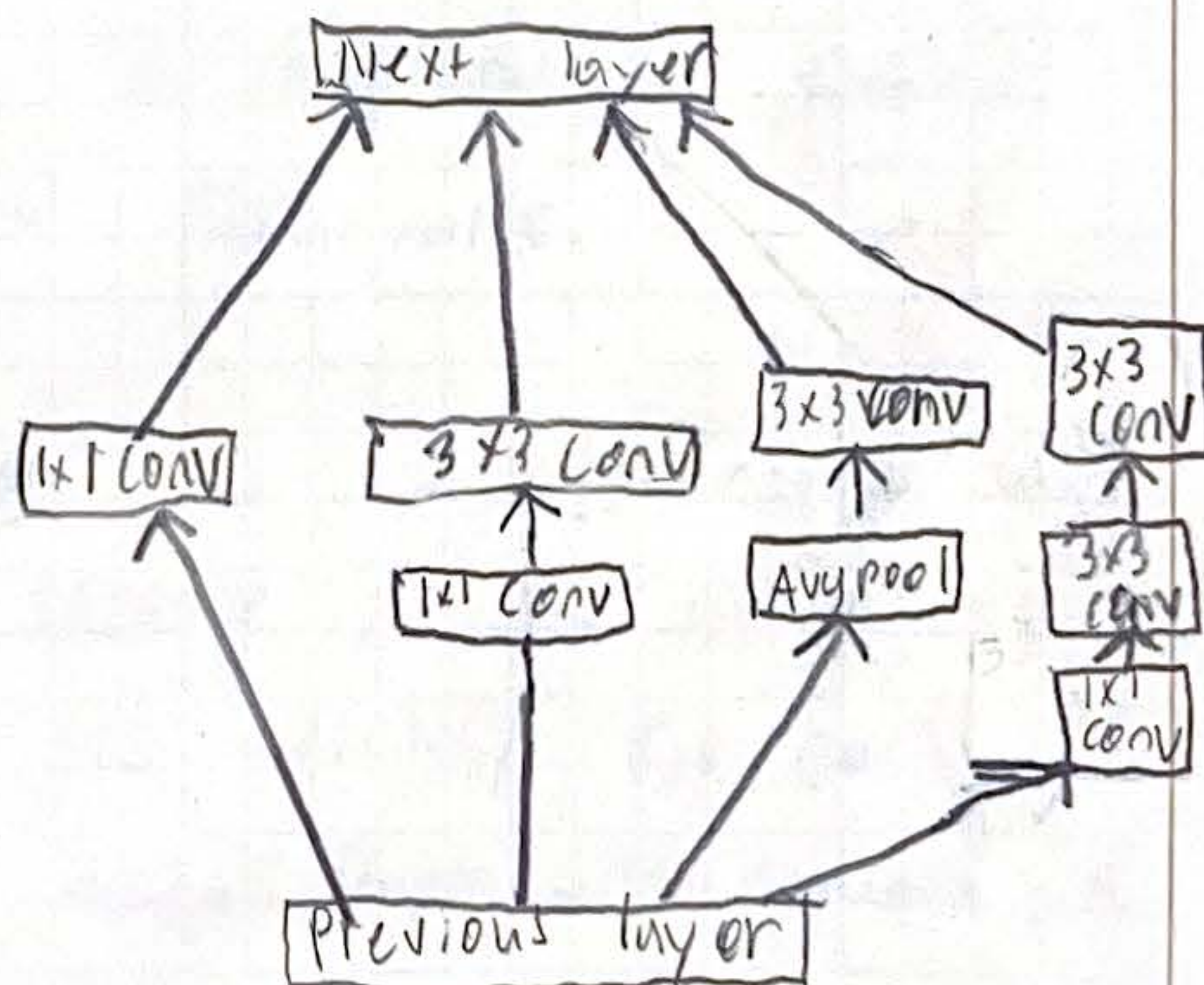
NOTE:  $\star$  These guys do not invent DSC  $\star$

Before we proceed we need to know what "Inception Net" is. It is a paper that came before this one and introduced the following idea: use filters of various sizes to Extract information at various scales and complexity.

old Networks:



(new)  
Inception v3:





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In this paper they discuss and show how a Depth wise - Seperable convolution is an extension of Inception net + gets better results for same # of params.

They start by noting that simple conv with more layers was state of the art + popular. Then based off of "Network in Network" paper

Inceptionet came + had better results. the Authors ask: How is it that mathimatically they are so similar + the Inceptionet works better? whats the next logical step?

The Hypothesis: (why Inception does better)

In normal convolution the kernel needs to map both cross-channel relationships and cross-spatial relationships.

AKA: I looks cross-channels first with  $1 \times 1$  conv and then it looks cross-spatial with  $3 \times 3$  conv. This decoupling is forced by the network structure.

After this they ask is it reasonable to these mappings totally separate....  $1 \times 1$  across all channels + then  $3 \times 3$  indivituly for each channel. This is pretty much Depth wise seperable conv.



They note the difference between  
"Extreme Inception" and already used "Depth wise  
separable conv":

1. In DSC the order is: spatial  $3 \times 3$  the channel wise  $1 \times 1$

In "Extreme" this is flipped

2. In DSC its  $\text{conv} \rightarrow \text{conv}$

In "Extreme" its  $\text{conv} \rightarrow \text{non-linear} \rightarrow \text{conv}$

#1 is irrelevant.

#2 is examined if it's important.

So they note that convolution lies on  
a spectrum where the two ends (DSC +  
normal convolution) correspond to # of  
channels used when doing spatial convolution

So they propose the Xception net  
architecture. A linear stack of DSC with  
residual connections.



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So they then do Experiments.

- the Xception net does Better than Inception. (no non-linearities in between  $1 \times 1$  &  $3 \times 3$ )
- Then they test if adding nonlinearity in between will help (Others used & found good results of doing so for Inception net NOTE: this builds off of the "Network in Network" paper)
  - They find NO improvement actually find it makes it worse.
  - Surprising Because It was helpful for others.
  - They think that the non-linearity is helpful in inception net cuz of the deep feature space
  - But in shallow feature spaces it leads to loss of info
  - Note: this is their hypo they don't expand.
- They note that the way they implemented DSC was slower than inception, But can be optimized on the Hardware in the future